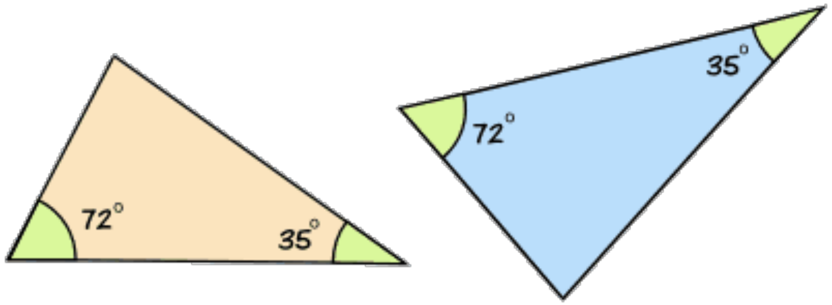
Similarity

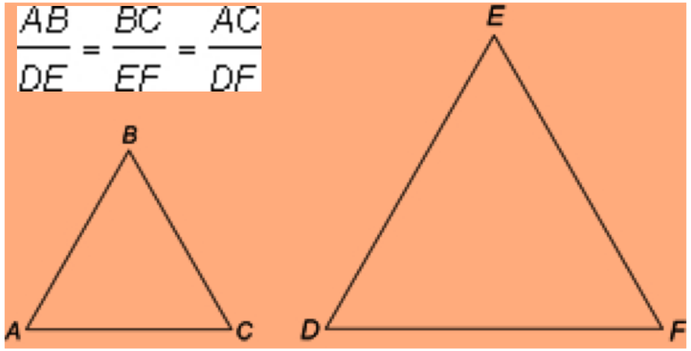
Lesson 3 – Similar Triangles

Similar Triangles

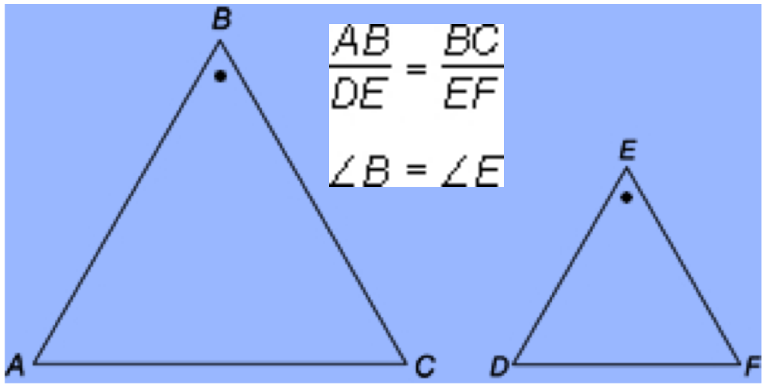
1. AAA



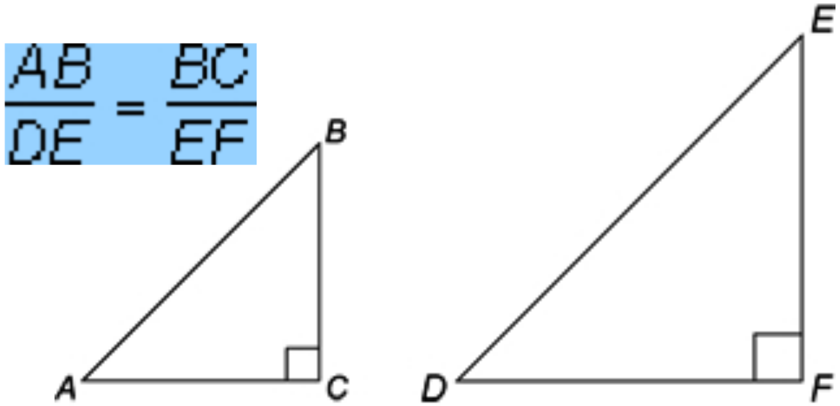
2. SSS



3. SAS



4. RHS



NOTATION

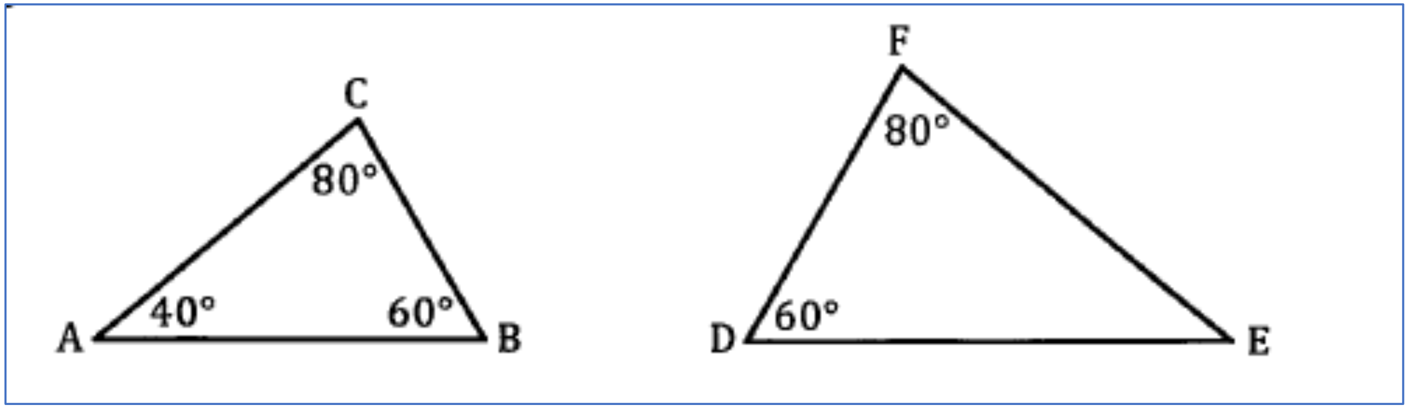
If triangle ABC is similar to triangle DEF, we can then write the following:

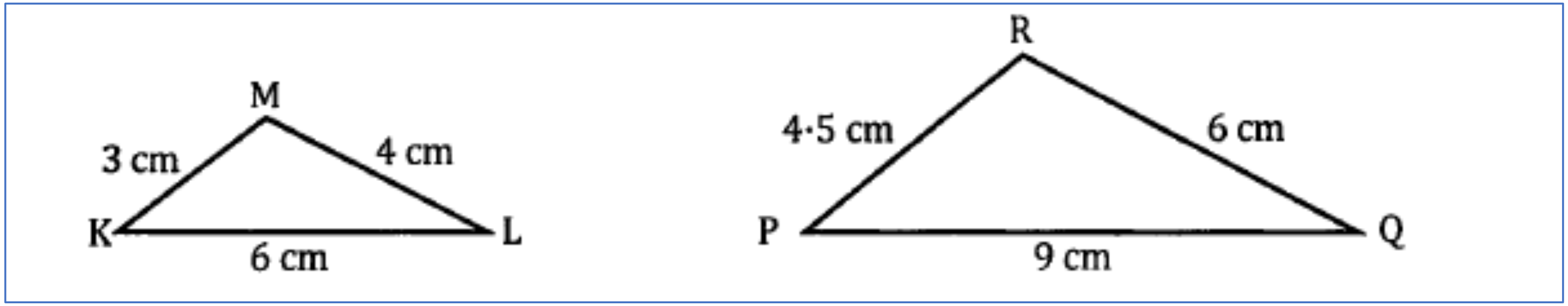
$$\boxed{\Delta\, ABC \sim \Delta\, DEF}$$

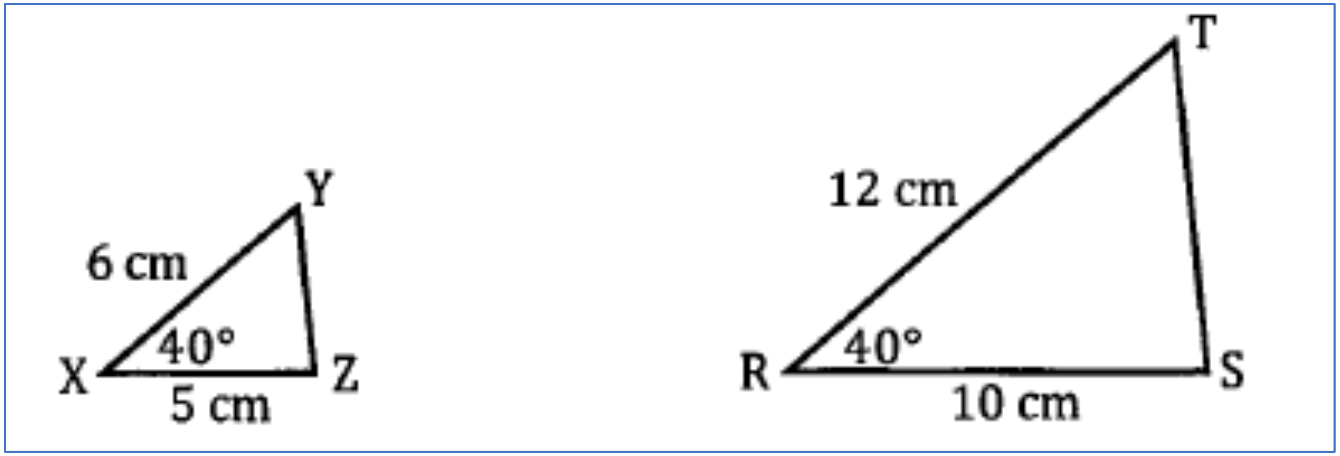
or

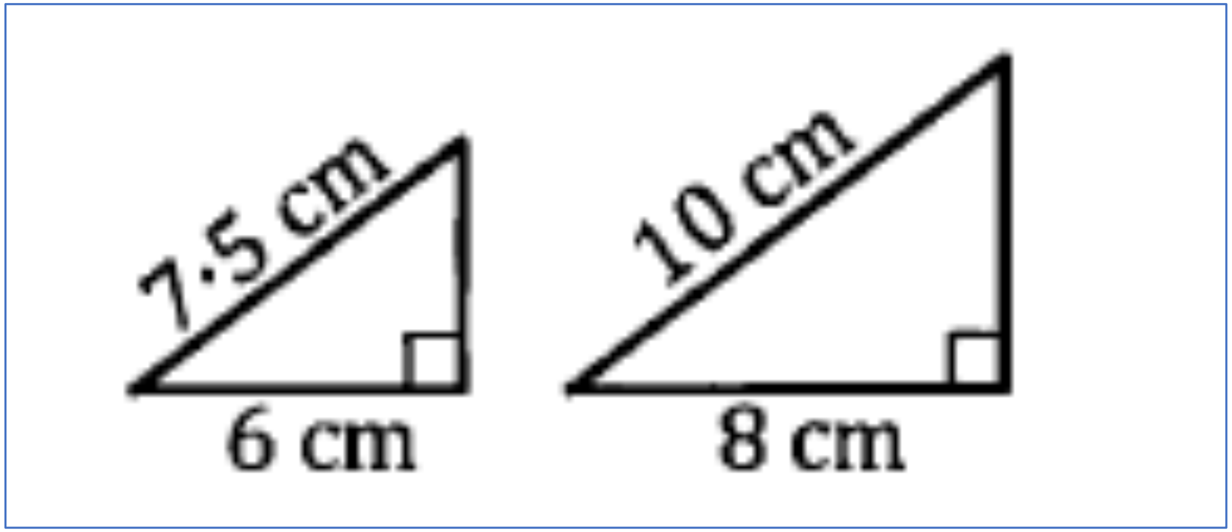
$$\Delta\, ABC \;|||\; \Delta\, DEF$$

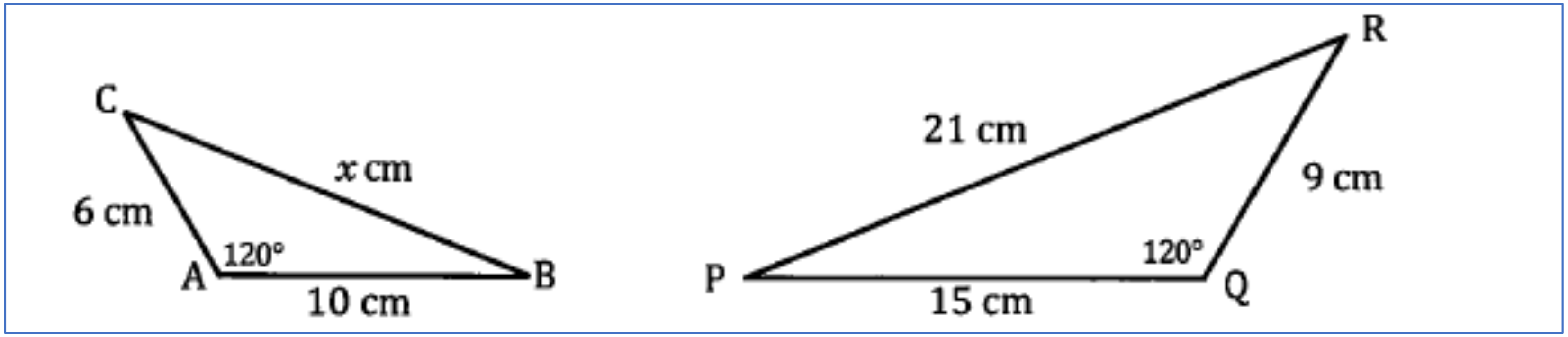
Examples






Y
6 cm
40°
X
Z
5 cm
T
12 cm
40°
R
S
10 cm


7·5 cm
6 cm
10 cm
8 cm



$\angle CAB = \angle PQR$

$\frac{QR}{AC} = \frac{9}{6} = \frac{3}{2}$

$\frac{QP}{AB} = \frac{15}{10} = \frac{3}{2}$

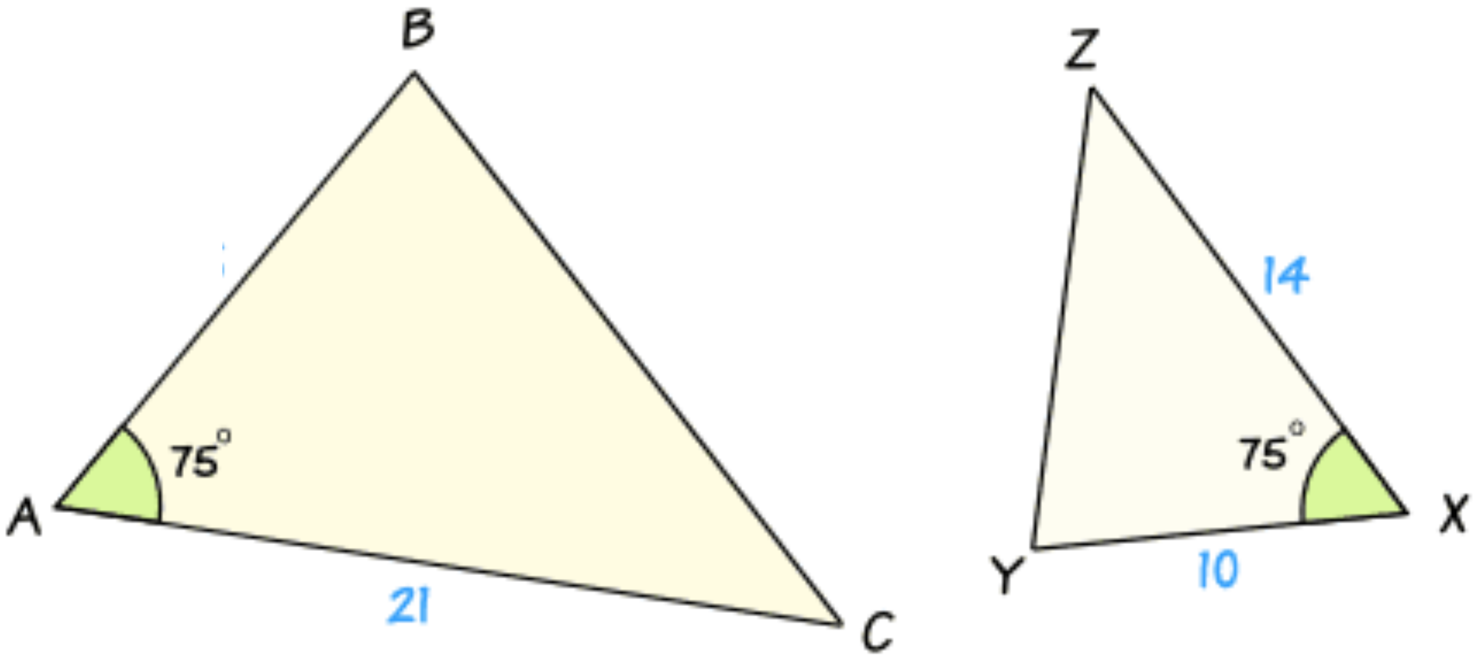
Therefore $\Delta ABC \sim \Delta QPR$ because two pairs of corresponding side is the same ratio and the included angles are equal

$\frac{RP}{CB} = \frac{21}{x} = \frac{3}{2}$

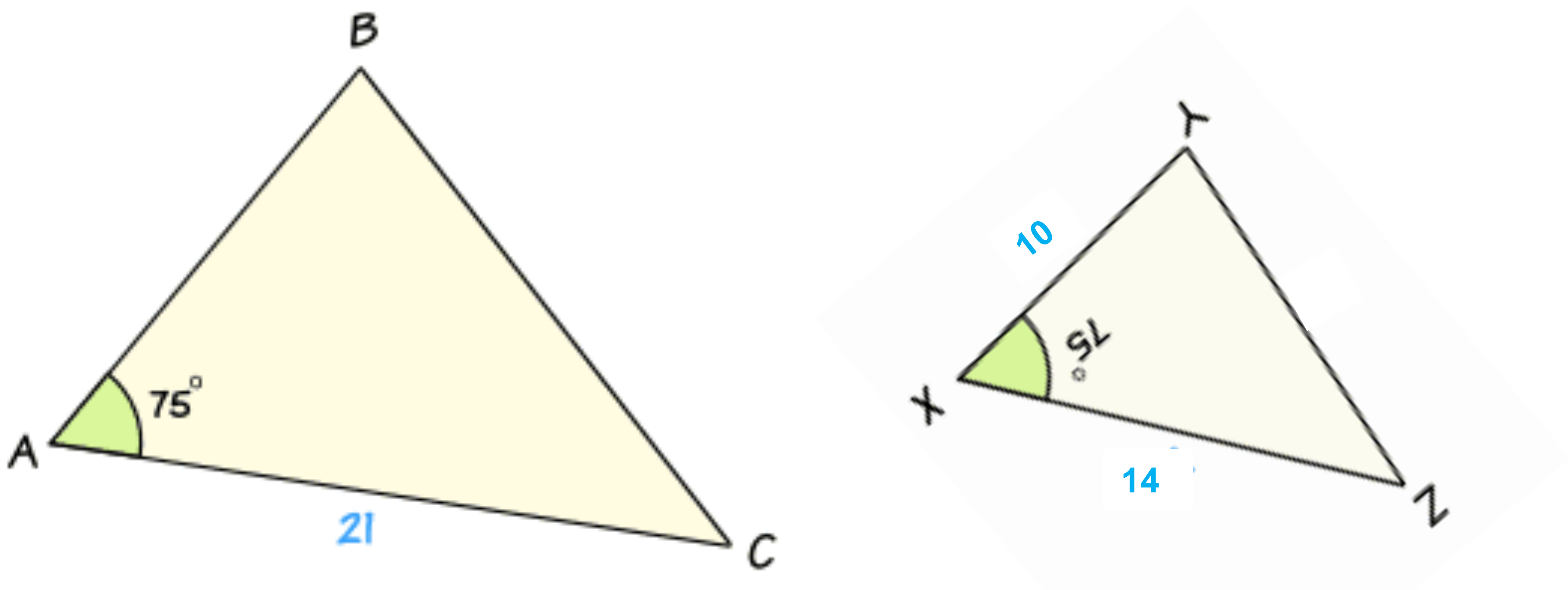
$x = \frac{2(21)}{3}$

$= 14$ cm

Example



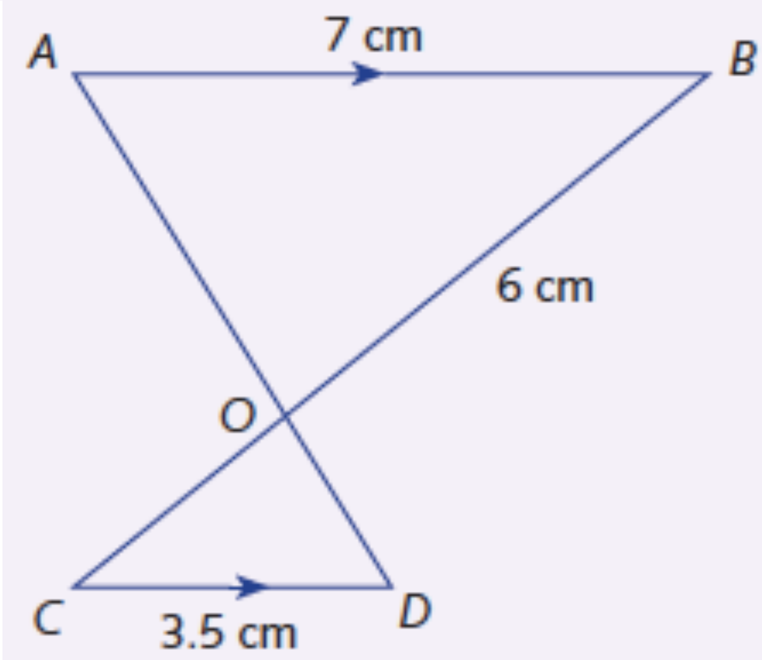
Given Δ ABC ~ Δ XYZ, find the length of AB.



1. Find the scale factor using the matching sides: Scale factor = $\frac{AC}{XZ} = \frac{21}{14} = \frac{3}{2}$
2. Find the length of AB by multiplying the scale factor to the matching side (in this case, XY): AB = $\frac{3}{2}$ x 10 = 15

Example 4

In the figure below, $AB \parallel CD$ and $AB = 7$ cm, $BO = 6$ cm and $CD = 3.5$ cm. Find CB.



Given $AB \parallel CD$

$\angle ABO = \angle DCO$ (alternate angles)

$\angle AOB = \angle DOC$ (vertically opposite angles)

$\angle BAO = \angle CDO$ (alternate angles)

Therefore $\Delta OAB \sim \Delta ODC$ (all corresponding angles equal)

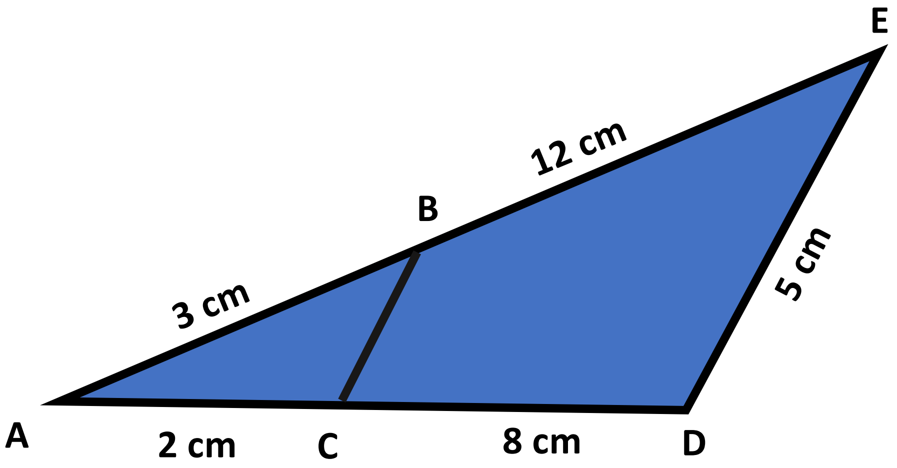
$$\frac{CD}{AB} = \frac{CO}{BO}$$

$$\frac{3.5}{7} = \frac{CO}{6}$$

$CO = 3$ cm

$CB = 6 + 3 = 9$ cm

Example 5



Find the length of BC given the BC is parallel to DE.

$\angle BAC = \angle EAD$ (common angle)

$\angle ABC = \angle AEC$ } corresponding angles

$\angle ACB = \angle ADE$ }

Therefore $\triangle ABC \sim \triangle AED$ because all corresponding angles are equal

$$\frac{BC}{DE} = \frac{3}{3+12}$$

$$\frac{BC}{5} = \frac{3}{15}$$

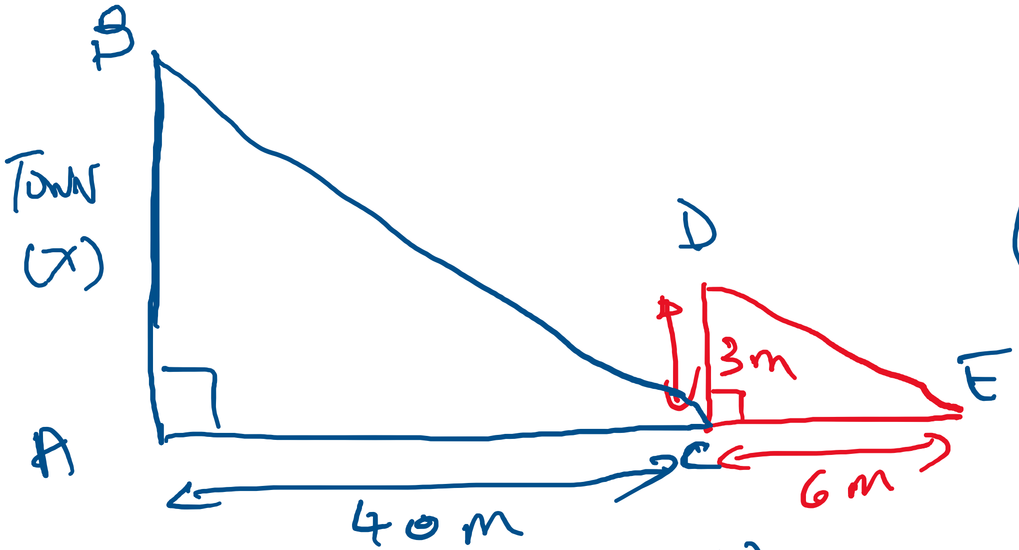
$$BC = 1 \text{ cm}$$

or
$$\frac{BC}{5} = \frac{2}{10}$$

$$BC = 1 \text{ cm}$$

Example 6

A tower standing on level ground casts a shadow 40 m long. A vertical stick 3 m high is placed at the tip of the shadow. The stick is found to cast a shadow 6 m long. Find the height of the tower.



$\angle BAC = \angle DCE$ (90°)

$BA \parallel DC$

$\angle BCA = \angle DEC$

$\triangle ABC \sim \triangle CDE$ (all corresponding angles equal)

$$\frac{40}{6} = \frac{x}{3}$$

$$x = \frac{3(40)}{\cancel{6}\,2} = 20$$

The tower is 20 m high.